

RNAbridge: a database of extended and non-canonical helices

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Introduction

Traditional RNA structural analysis often overlooks the **extended non-canonical helices** that form the core of complex 3D architectures. These motifs, consisting of canonical stems joined by non-canonically interconnected bulges or internal loops, are essential for understanding RNA folding and function.

These helices are foundational for accurate RNA 3D structure prediction, structural validation, and deciphering the spatial organization of complex junctions.

RNAbridge [1] is the first resource that systematically catalogs these building blocks, bridging the gap between 2D topology and 3D spatial organization.

What are extended non-canonical helices?

An extended non-canonical helix is a structural motif in which non-canonical connectors bridge canonical stems while maintaining a continuous stacking environment. These helices often form the structural scaffolding of complex multi-way junctions or originate from them.

RNAbridge extracts these motifs from RNA 3D structures by analyzing base-pairing and stacking interactions.

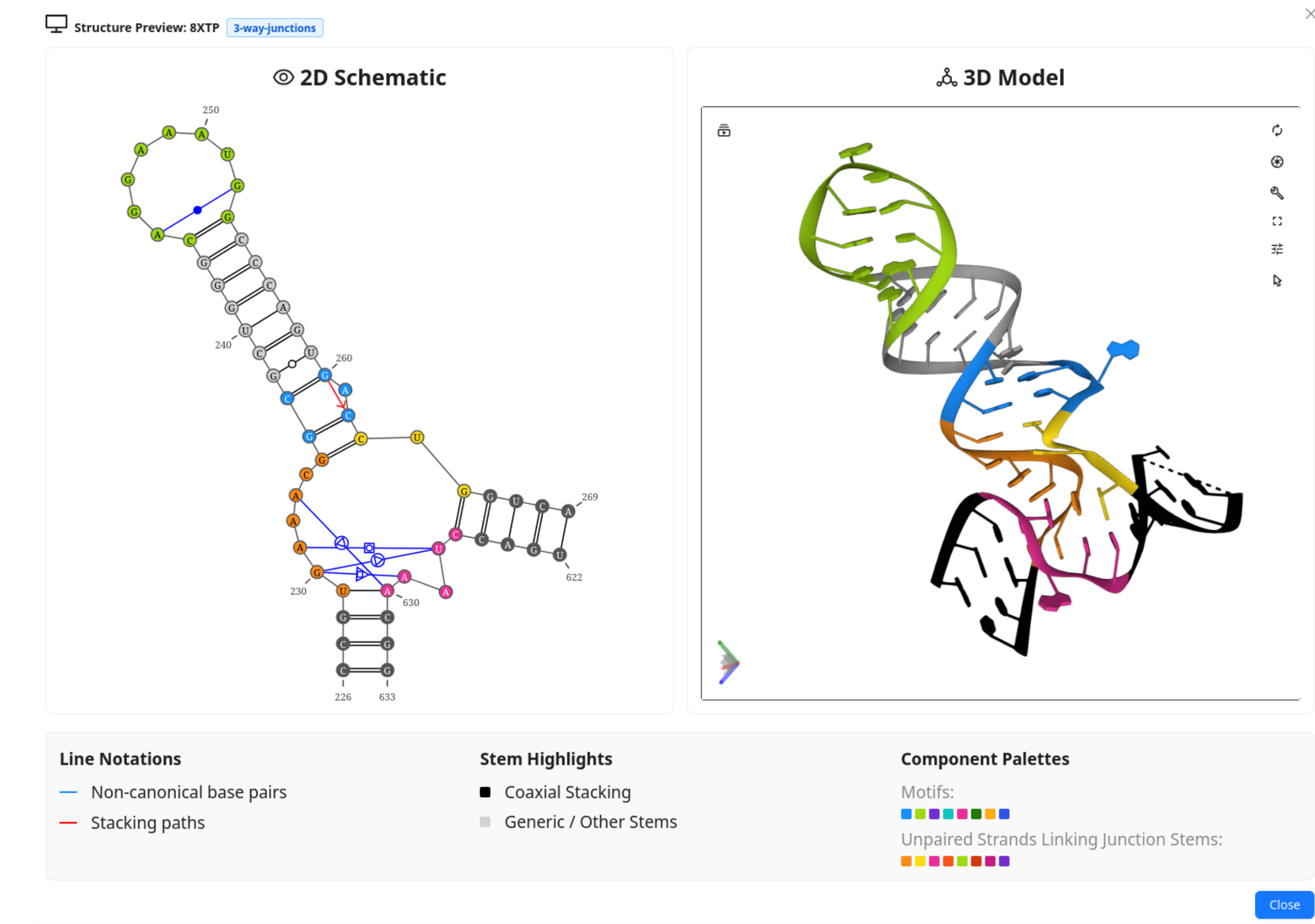


Fig 1. Dual representation of an extended non-canonical helix (*Comamonas testosteroni* KF-1 circularly permuted group II intron Post-2S state, PDB ID: 8XTP): 2D topology (left) and its corresponding 3D architecture (right).

Key concepts for motif identification

Identification of coaxial stacking relies on precise geometric and topological measurements calculated within the RNAbridge engine, utilizing **X3DNA** [2] for structural analysis:

- **Stacking Paths:** Verified using a graph-based BFS algorithm within the core structural engine.
- **Global Bend Angle:** Measured between the first and the last canonical stems.
- **Local Bend Angle:** Measured between adjacent stems within the motif.
- **Sequence Continuity:** Preventing gaps in the sequence.

Helix Components						Download JSON
ID	Type	Size (2D / 3D)	Local Angle	Stacking Path	Sequence	
1	STEM	3x3	-	-	GGG-CCC	
2	BULGE-OUT	1x0 0x0	35.4°	A126-A128	A	
3	STEM	2x2	-	-	CG-CG	
4	INTERNAL_LOOP	2x2 2x2	21.6°	1: A101...A104 2: A125...A122	CU-UU	
5	STEM	6x6	-	-	GGUGAG-CUCACC	
6	HAIRPIN	7 nt	-	-	UGCAACU	

Fig 2. Motif examples characteristics.

RNAbridge Workflow

1. **Data Input & Pre-processing:** Normalize CIF files, fetch experimental metadata from the RCSB PDB, and perform automated annotation.
2. **Interaction Mapping:** Construct high-speed lookup indices for all canonical and non-canonical interactions.
3. **Motif Classification:** Identify and label structural elements, such as hairpins, bulges, and N-way junctions.
4. **Structural Analysis & Quantification:** Verify coaxial stacking stability using BFS pathfinding and measure 3D helical bend angles with X3DNA [2].
5. **Helical Assembly:** Recursively group motifs into global super-helices based on spatial orientation and geometric distances.
6. **Classification & Visualization:** Cluster results taxonomically and generate interactive 2D (SVG) and 3D (PML/CIF) visualizations.
7. **Data Integration & Persistence:** Map biological metadata and hierarchical findings into a searchable relational database.

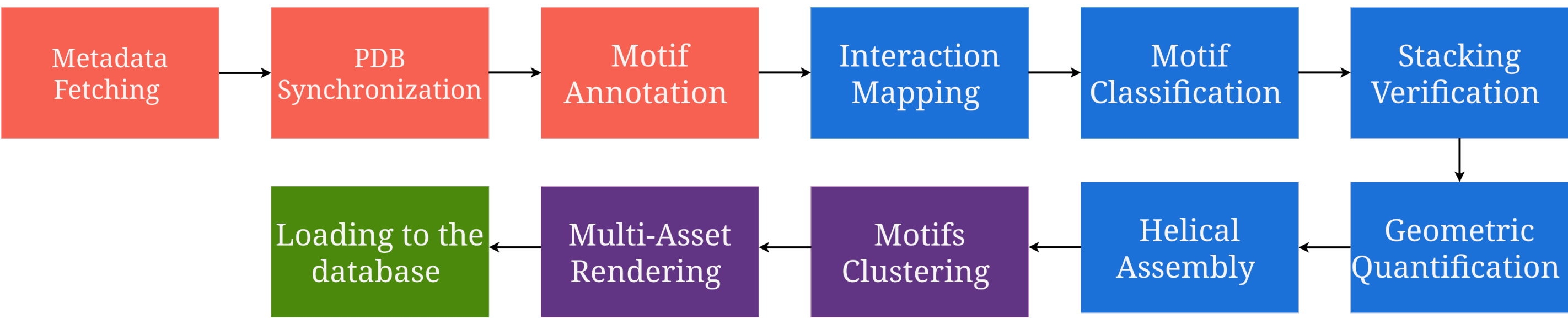


Fig 3. RNAbridge extraction pipeline.

Quantitative Analysis

RNAbridge currently catalogs a comprehensive collection of **75,004 non-canonical structural motifs**. The following histogram shows the distribution of these motifs across structural categories based on bend angles.

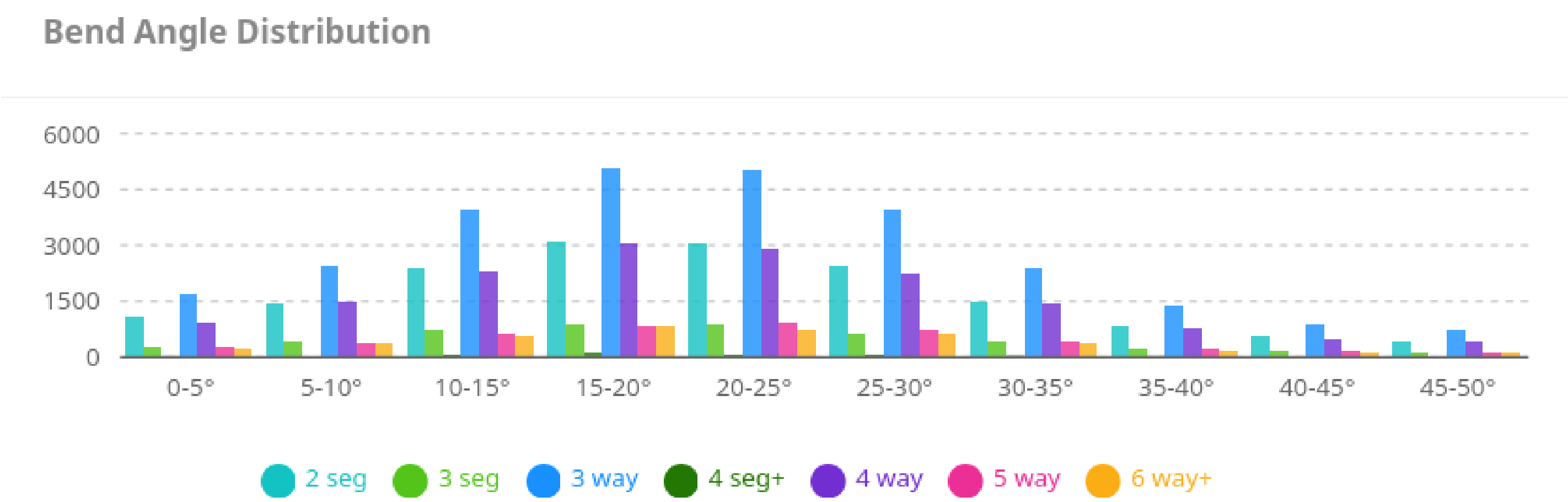


Fig 4. Bend-angle based distribution for the selected motif categories (2-, 3- and 4+ seg helices; and 3-, 4-, 5- and 6+- way junctions).

Helices	Count	Junctions	Count
2-segment	16,835	3-way	27,634
3-segment	4,813	4-way	16,156
4+ segment	555	5-way	4,805
		6+ way	4,206
Total			75,004

Tab 1. Quantitative summary of identified non-canonical structural motifs.

Web Server: rnabridge.cs.put.poznan.pl | Source Code: github.com/onidd/RNABridge

Bibliography

- [1] Zakrzewski D., Antczak M., Zok T. **RNAbridge: a database of extended and non-canonical helices.** *In preparation.*
- [2] Lu, X. J., & Olson, W. K. (2003). 3DNA: a software package for the analysis, rebuilding and visualization of three-dimensional nucleic acid structures. *Nucleic acids research*, 31(17), 5108-5121

Acknowledgements

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